# IP Addresses and Windows Command Prompt Networking Tools

## IP Addresses

Watch the video https://www.youtube.com/watch?v=ThdO9beHhpA and answer the following questions:

1. What is an IP address?
   A unique address that identifies each computer on a network.
2. What does IP stand for?
   Internet protocol
3. How many parts are there in an IP address?
   2 parts,network id and host id
4. List and describe these parts.
   8 of the bits is variable and represents the device on the network.The other 3 bytes represent the network ID
5. There are two types of IP addresses. What are they?
   IP version4 andIP version 6
6. What does an IPv4 address look like?
   255.255.255.255
7. How many bits are used in this address?
   32
8. What name is given to each number in an IPv4 address?
   octet
9. What is the number range allowed for each number in the IPv4 address?
   0-255
10. Approximately how many IPv4 addresses are there?
    Around 4 billion (2^32)
11. How would a computer view the IP address 1.2.3.4?
    As a valid IP address
12. What is the largest decimal number allowed for each octet in an IP address?
    255
13. What is the smallest number allowed for each octet in the address?
    0
14. Why was IPv6 created?
    There are more and more computers connected to the internet,reaching close to the 4 Billion limit,meaning that a new standard was needed
15. How long is this address in terms of the number of bits?
    128 bits
16. How many IP addresses are allowed under IPv6?
    $2^{128}$ (an extremely large number)
18. How many bits are used for each hex digit in an IPv6 address?
    4 bits
19. How many total possible hex digits are found in an IPv6 address?
    32 hex digits

## Windows Networking Tools

### PING

Watch the video https://www.youtube.com/watch?v=IlicPE38O-s to answer the following questions:

1. What is the purpose of this tool? Test connectivity between your computer and another device or server.
2. What do you think it means by testing for DNS name resolution?Converts a website name (e.g., google.com) into its IP address.

3. What information is required to be used with the ping command?

   An IP address or hostname

4. Write an example command using ping.
   ping google.com
5. How many packets of data are sent with every ping command?

   By default, 4 packets (on Windows)

6. How many packets should be received from the computer that sent out the ping command?

   Should also be 4 if all packets are received

7. If the same number of packets sent matches the number received what could we logically conclude from this?

   Connection is successful; no packet loss

8. How could having a firewall affect the response to a ping command?

   May block ICMP requests, causing packet loss or no response

9. What is one reason why a computer may not get all the packets back that it sent when using the ping command?

   Network congestion, firewall, or device restrictions

10. How could I use the ping command to test connectivity to a website?
    ping google.com
11. What is the loopback address and what does it represent?

    127.0.0.1 – Tests if the local computer's networking stack works

12. How does pinging this address help you?

    Confirms your own network interface is working

13. What does DNS stand for and what is it used for?

    Domain Name System – translates domain names into IP addresses

14. What is google's DNS server address?

    8.8.8.8

15. What are command switches?

    Modify the behavior of a command

16. Provide an example of a switch used with ping.
    ping -t google.com
17. Do some research and list 2 other switches and what they do.

18. Using your practice computer access the windows console. Ping the loopback address. Insert a screenshot of the response.

19. Ping the address of your router. Insert a screenshot of the response.
20. Ping 1.2.3.4. Insert a screenshot of the response. Explain what you see.
21. Try the following: ping -a 127.0.0.1

## IPCONFIG

Watch the video https://www.youtube.com/watch?v=ZKhorleA5aA to answer the following questions:

1. What is the purpose of this tool?

   Displays current TCP/IP network configuration

2. What are 4 basic pieces of info this command can provide?

   IPv4 Address

   IPv6 Address

   Subnet Mask

   Default Gateway

3. What is the subnet mask used for?

   Defines which part of the IP address is network vs host

4. What is the Default Gateway?

   Device that forwards traffic to other networks (e.g., your router)

5. Why do most modern computers have both an IPv4 and an IPv6 address?

   For compatibility with both old and new networks

6. The IPv4 address listed is a private or public IP address?

   Usually private (like 192.168.x.x)

7. How can you find your public IP address?

   Search “what is my IP” in a browser

8. Provide an example switch used with this command and explain what it does.

   Shows detailed information including MAC address, DHCP server

9. List 2 additional pieces of info this switch provides.

   Lease times, DNS servers

10. What is another name for the Physical Address listed under ipconfig /all?

    Unique identifier for your network adapter

11. What is the Physical Address representing?

    Assigns IP addresses automatically

12. What does DHCP stand for?
13. What does DHCP mean?
14. What is a DHCP server?

    Device that assigns IP addresses

15. What is a DNS server?

    Translates domain names to IP addresses

16. What does the command ipconfig /flushdns do?

    Clears DNS resolver cache

17. What is DNS Spoofing and why is performing a DNS flush a good counter to this?
18. What does the ipconfig /displaydns command do?
19. What switch can you use to find more info on other available switches?
20. Access your practice computer and find your attached routers IP address. Insert a screenshot here.
21. Access your practice computer and find out its MAC address.
22. Embed a screenshot of your computer's DNS resolver cache.
23. What is the IP address of your DHCP server? Explain why this might be the same as your Default Gateway.